

Oefeningenreeks 1

1. a.

$$\{0^n 1^n \mid n \in \mathbb{N}_0\} = \{01, 0011, 000111, 00001111, \dots\}$$

$$P = S \rightarrow 0S1 \mid 01$$

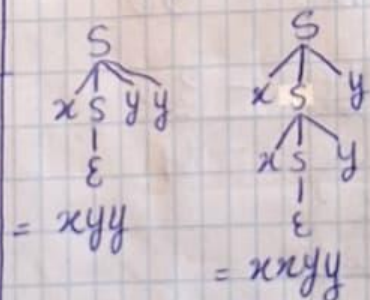
$$G = (\{S\}, \{0, 1\}, P, S)$$

b.

$$\{x^n y^m \mid m, n \in \mathbb{N} \text{ met } m=n, \text{ of } m=2n\}$$

$$= \{\epsilon, xy, xyy, xxxy, xxxxy, xxxxyy, xxxxyyy, \dots\}$$

$$P = S \rightarrow xSy \mid xSyy \mid \epsilon$$



$$G = (\{S\}, \{x, y\}, P, S)$$

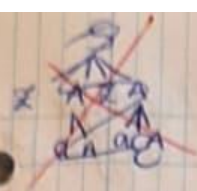
c.

~~$$R = \{a^* a^n b^m a^m b^* \mid n, m \in \mathbb{N}\}$$~~

~~$$S \rightarrow AABAB$$~~

~~$$A \rightarrow aA \mid \epsilon$$~~

~~$$B \rightarrow bA \mid \epsilon$$~~



11. $P =$
 $S \rightarrow \epsilon AB \epsilon$
 $A \rightarrow aAa \mid \epsilon$
 $B \rightarrow bBb \mid \epsilon$

$$G = (\{S, A, B\}, \{a, b, \epsilon\}, P, S)$$

d. $\{a^i b^j c^k \mid i, j, k \in \mathbb{N}, i \neq j \text{ of } j \neq k\}$

$= \{b, ac, \dots\}$

2.

* Linkse afleiding:

$$\begin{aligned} S &\Rightarrow A 1 B \\ &\Rightarrow 0A 1 B \\ &\Rightarrow 00A 1 B \\ &\Rightarrow 00\epsilon 1 B \\ &\Rightarrow 0010B \\ &\Rightarrow 00101B \\ &\Rightarrow 00101\epsilon \\ &\Rightarrow 00101 \end{aligned}$$

* Rechtse afleiding

$$\begin{aligned} S &\Rightarrow A 1 B \\ &\Rightarrow A 1 0B \\ &\Rightarrow A 1 01B \\ &\Rightarrow A 1 01\epsilon \\ &\Rightarrow 0A 1 01 \\ &\Rightarrow 00A 1 01 \\ &\Rightarrow 00\epsilon 1 01 \Rightarrow 00101 \end{aligned}$$

3. $P =$
 $\{ S \rightarrow ACbB, \\ A \rightarrow aA \mid \epsilon, \\ B \rightarrow bB \mid \epsilon, \\ C \rightarrow bC \mid cC \mid \epsilon \}$

$$G = (\{S, A, B, C\}, \{\epsilon, a, b, c\}, P, S)$$

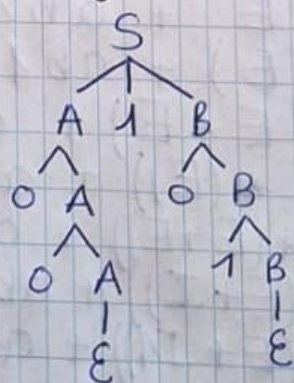
4.

$$\begin{aligned} P = \\ \{ S &\rightarrow S + S \\ S &\rightarrow S * S \\ S &\rightarrow (S) \\ S &\rightarrow 0S \\ S &\rightarrow 1S \\ S &\rightarrow e \\ S &\rightarrow \emptyset \} \end{aligned}$$

$$G = (\{S\}, T, P, S)$$

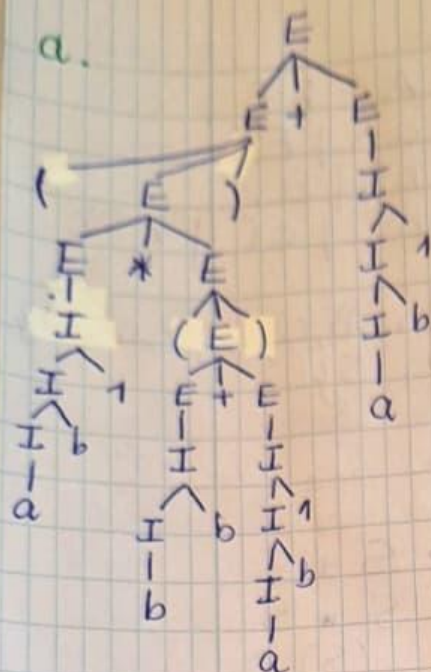
Oefeningen reeks 2

1.



2.

a.



$$= (ab1 * (bb + ab1)) + ab1$$

b. Rechte afleiding

$$\begin{aligned}
 E &\Rightarrow E + E \\
 &\Rightarrow E + I \\
 &\Rightarrow E + I1 \\
 &\Rightarrow E + Ib1 \\
 &\Rightarrow E + ab1 \\
 &\Rightarrow (E) + ab1 \\
 &\Rightarrow (E * E) + ab1 \\
 &\Rightarrow (E * (E)) + ab1 \\
 &\Rightarrow (E * (E + E)) + ab1 \\
 &\Rightarrow (E * (E + I)) + ab1 \\
 &\Rightarrow (E * (E + I1)) + ab1 \\
 &\Rightarrow (E * (E + ab1)) + ab1 \\
 &\Rightarrow (E * (I + ab1)) + ab1 \\
 &\Rightarrow (E * (Ib + ab1)) + ab1 \\
 &\Rightarrow (E * (bb + ab1)) + ab1 \\
 &\Rightarrow (I * (bb + ab1)) + ab1 \\
 &\Rightarrow (I1 * (bb + ab1)) + ab1
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow (Ib1 * (bb + ab1)) + ab1 \\
 &\Rightarrow (ab1 * (bb + ab1)) + ab1
 \end{aligned}$$

c.

$$\begin{aligned}
 E &\Rightarrow E + E \\
 &\Rightarrow (E) + I \\
 &\Rightarrow (E * E) + I1 \\
 &\Rightarrow (I * (E)) + ab1 \\
 &\Rightarrow (ab1 * (E + E)) + ab1 \\
 &\Rightarrow (ab1 * (I + I)) + ab1 \\
 &\Rightarrow (ab1 * (bb + ab1)) + ab1
 \end{aligned}$$

3.

$$a. ((aa * bb) + a1b) *$$

$$(a * (b0 + b))$$

b. Linkse afleiding

$$\begin{aligned}
 E &\Rightarrow E * E \\
 &\Rightarrow (E) * E \\
 &\Rightarrow (E + E) * E \\
 &\Rightarrow ((E) + E) * E \\
 &\Rightarrow ((E * E) + E) * E \\
 &\Rightarrow ((I * E) + E) * E \\
 &\Rightarrow ((Ia * E) + E) * E \\
 &\Rightarrow ((aa * I) + E) * E \\
 &\Rightarrow ((aa * bb) + Ib) * E \\
 &\Rightarrow ((aa * bb) + a1b) * E \\
 &\Rightarrow ((aa * bb) + a1b) * (E)
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow ((aa * bb) + a1b) * (E * E) \\
 &\Rightarrow ((aa * bb) + a1b) * (I * E) \\
 &\Rightarrow ((aa * bb) + a1b) * (a * (E)) \\
 &\Rightarrow " * (a * (E + E)) \\
 &\Rightarrow " * (a * (I + E)) \\
 &\Rightarrow " * (a * (I + E)) \\
 &\Rightarrow " * (a * (I + E)) \\
 &\Rightarrow " * (a * (b0 + E)) \\
 &\Rightarrow " * (a * (b0 + I)) \\
 &\Rightarrow " * (a * (b0 + b))
 \end{aligned}$$

$$E \xrightarrow{*} ((aa * bb) + a1b) * (a * (b0 + b))$$

Oefeningen reeks 3

1.

* Voor een string met ^{de} lengte 0 hebben we:

$$B \rightarrow (B) \rightarrow (\epsilon)$$

Dus deze grammatica genereert een string, waarvan de lengte gelijk is aan 0 die bestaat uit gebalanceerde haakjes en .

* We veronderstellen dat

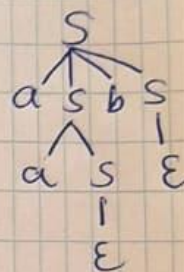
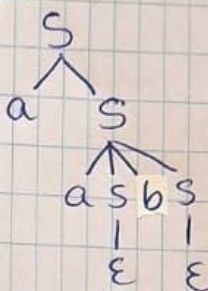
~~V string met lengte n~~
we ~~kunnen~~ een string, bestaande uit gebalanceerde haakjes kunnen genereren waarvan lengte gelijk is aan n.

2.

$$\begin{aligned}
 A &\rightarrow (A) \\
 A &\rightarrow \{ \} A \\
 A &\rightarrow [A] \\
 A &\rightarrow [\] A \\
 A &\rightarrow () []
 \end{aligned}$$

3.

a.



b.

$$\begin{aligned}
 S &\Rightarrow aSbS \\
 &\Rightarrow aaSbS \\
 &\Rightarrow aa bS \\
 &\Rightarrow aab
 \end{aligned}$$

$$\begin{aligned}
 S &\Rightarrow aS \\
 &\Rightarrow aaSbS \\
 &\Rightarrow aa bS \\
 &\Rightarrow aab
 \end{aligned}$$

c.

$$\begin{aligned}
 S &\Rightarrow aSbS \\
 &\Rightarrow aSb \\
 &\Rightarrow aaSb \\
 &\Rightarrow aab
 \end{aligned}$$

$$\begin{aligned}
 S &\Rightarrow aS \Rightarrow aaSbS \\
 &\Rightarrow aaSb \Rightarrow aab
 \end{aligned}$$

* Oefeningenreeks 3

1.

$$\text{base } * |w| = 0$$

$$w = \varepsilon$$

$$\text{Dus } B \Rightarrow \varepsilon$$

$$* |w| = 2$$

$$B \Rightarrow (B) \Rightarrow ()$$

IH:

Alle strings van lengte $|w| = 2n$ zijn gebalanceerd

$$|w| = 2(n+1) = 2n+2$$

$$\bullet w = (w_1)$$

$$B \Rightarrow (B) \xrightarrow[\text{IH}]{*} (w_1)$$

$$\bullet w = w_1 w_2 \text{ met } m_1, m_2 \neq \varepsilon$$

$$B \Rightarrow BB \xrightarrow[\text{IH}]{*} w_1 w_2$$